

Lab 1B: Regular Expressions

A *regular expression* is compact notation for a language (that is, a set of strings). Formally, a regular language is one of the following:

- The symbol $\emptyset$ (representing the empty set)
- Any string (representing the set containing only that string)
- $R + S$ for some regular expressions R and S (representing alternation / union)
- $R \cdot S$ or RS for some regular expressions R and S (representing concatenation)
- R^* for some regular expression R (representing Kleene closure / unbounded repetition)

In the absence of parentheses, Kleene closure has highest precedence, followed by concatenation. For example:

$$\begin{aligned}0 + RE^* &= 0 + R(E^*) = \{0, R, RE, REE, REEE, REEEE, REEEEE, REEEEE, REEEEE, \dots\} \\0 + (RE)^* &= \{\varepsilon, 0, RE, RERE, RERERE, RERERERE, RERERERERE, RERERERERE, \dots\} \\(0 + R)E^* &= \{0, R, 0E, RE, OEE, REE, OEEE, REEE, OEEEE, REEEE, OEEEE, \dots\} \\(0 + RE)^* &= \{\varepsilon, 0, 00, RE, 000, 0RE, RE0, 0000, 00RE, 0RE0, RE00, RERE, \dots\}\end{aligned}$$

Give regular expressions for each of the following languages over the binary alphabet $\{0, 1\}$. (For extra practice, find multiple regular expressions for each language.)

0. All strings.

$$(0 + 1)^*$$

1. All strings containing the substring 000 .

$$((0 + 1)^* 000 (0 + 1)^*)$$

2. All strings *not* containing the substring 000 .

$$(1 + 01 + 001)^*(\varepsilon + 0 + 00)$$

3. All strings in which every run of 0 s has length at least 3.
4. All strings in which the *last* 1 appears before the *first* substring 000 .
5. All strings containing at least three 0 s.
6. Every string except 000 . [Hint: Don't try to be clever.]